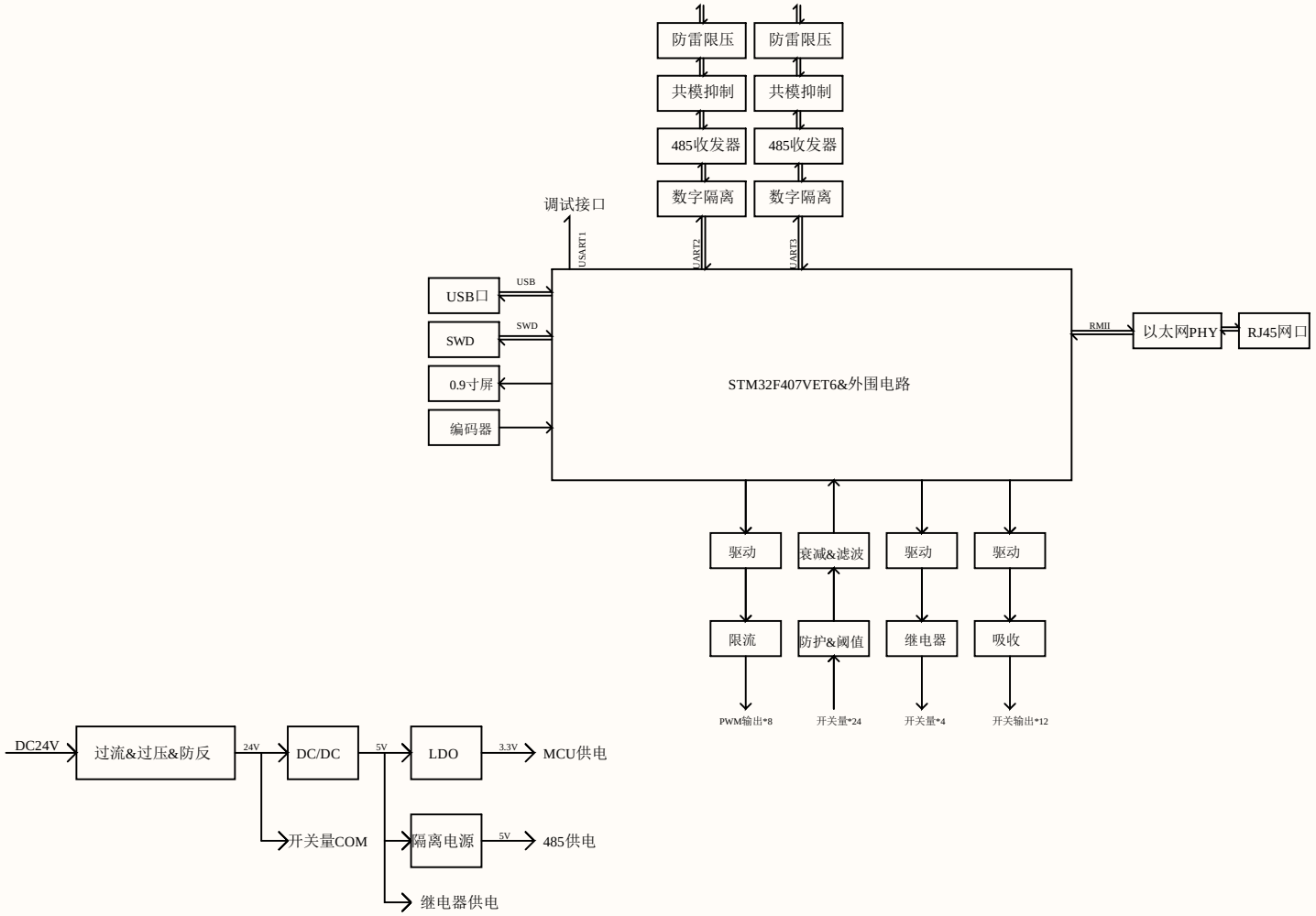
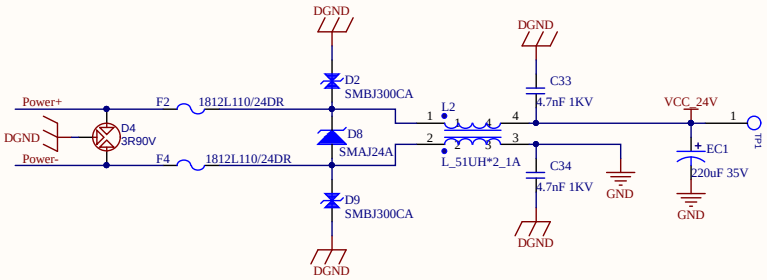
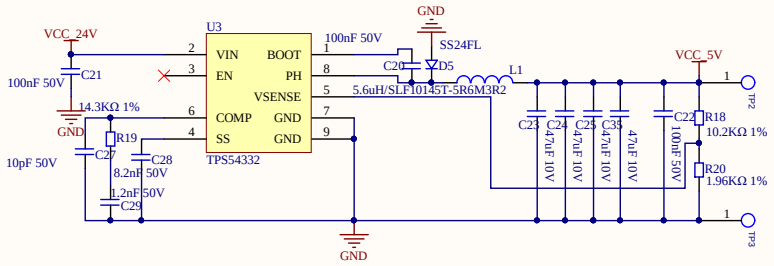
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| V1.1 | 2023.08.15 | 桑牧丹 | 1、Q1，三极管封装错误 |

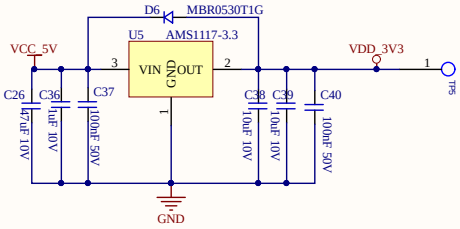
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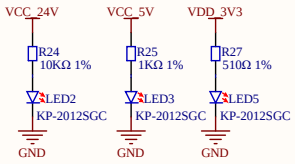



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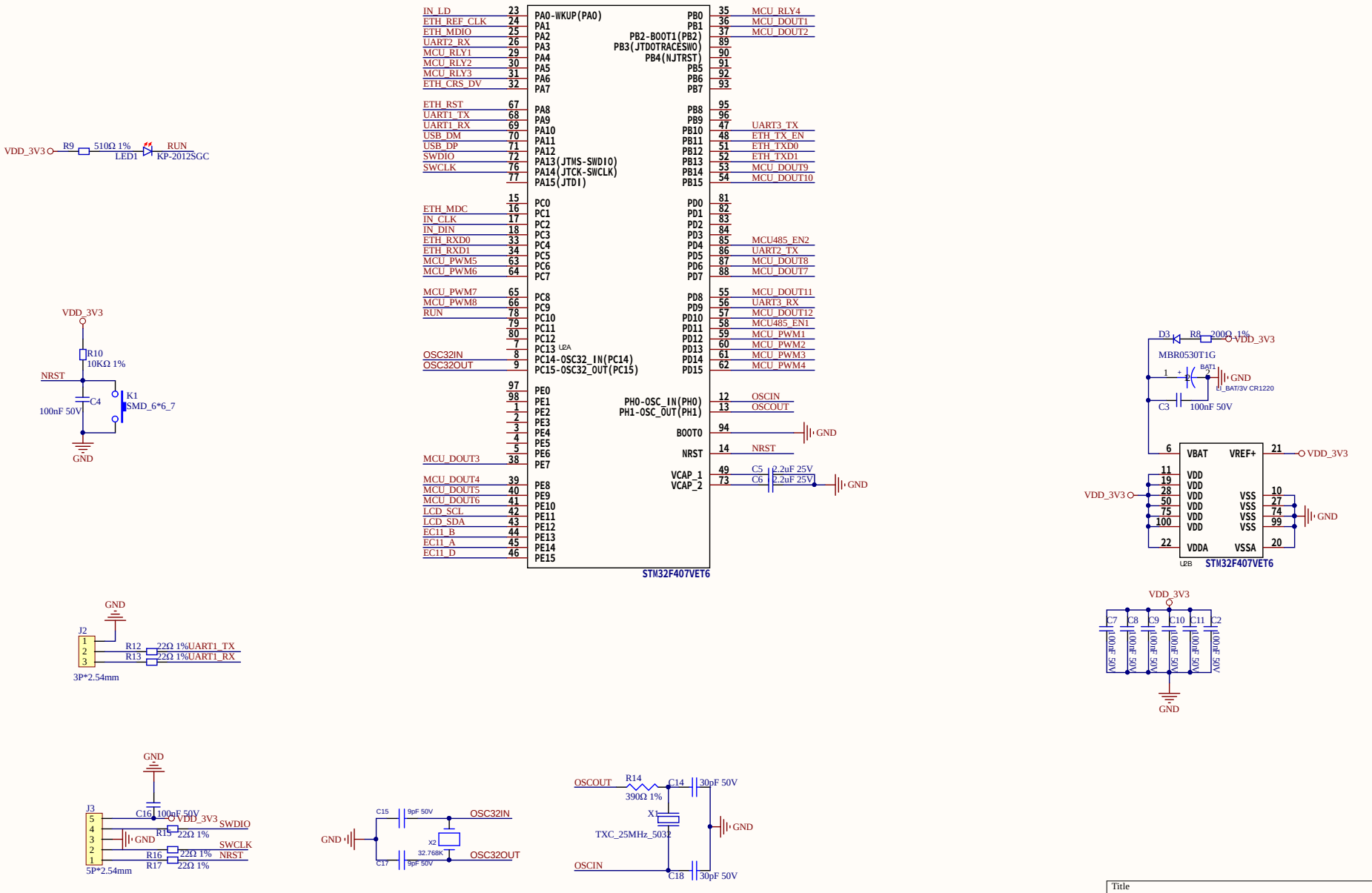



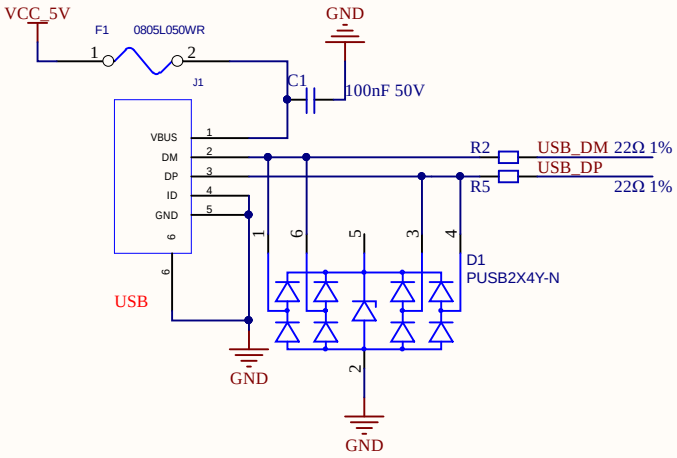




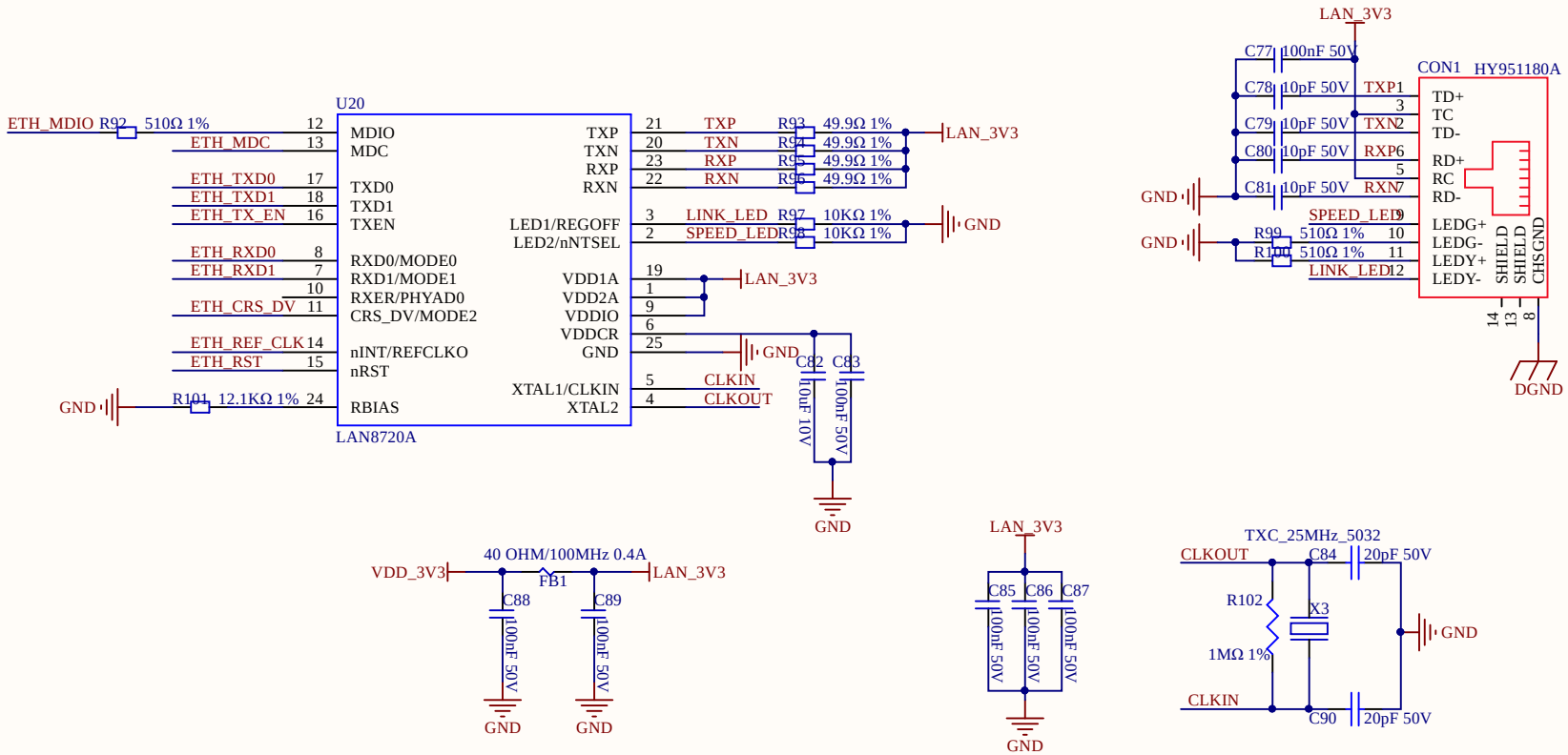
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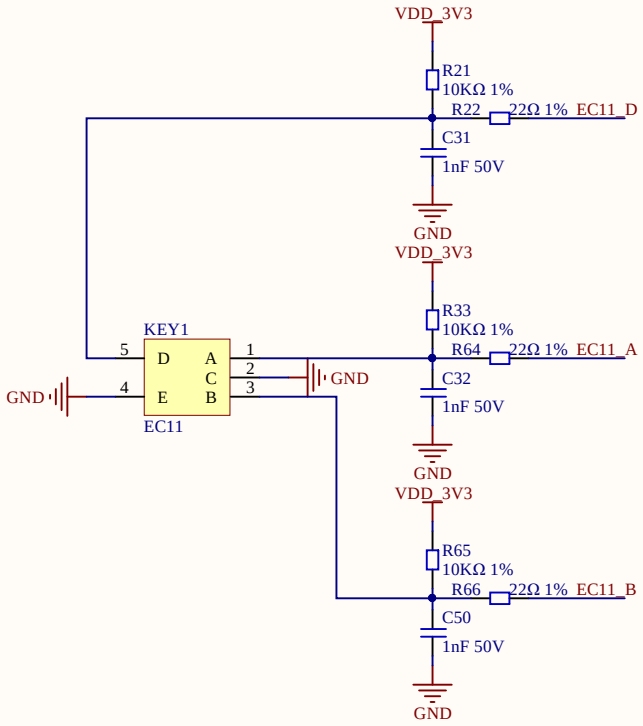

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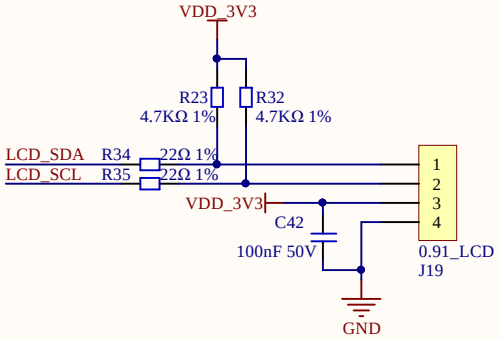



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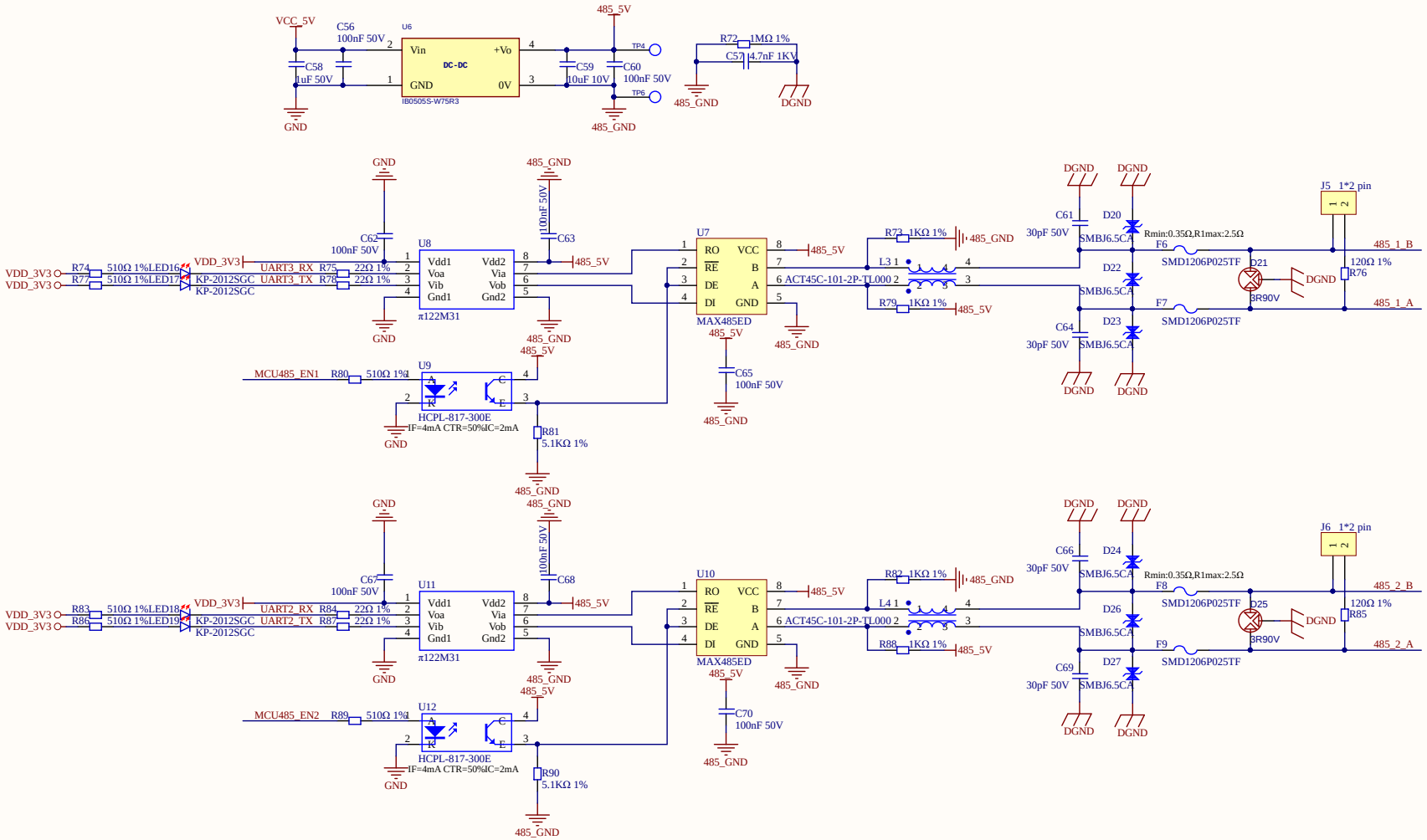



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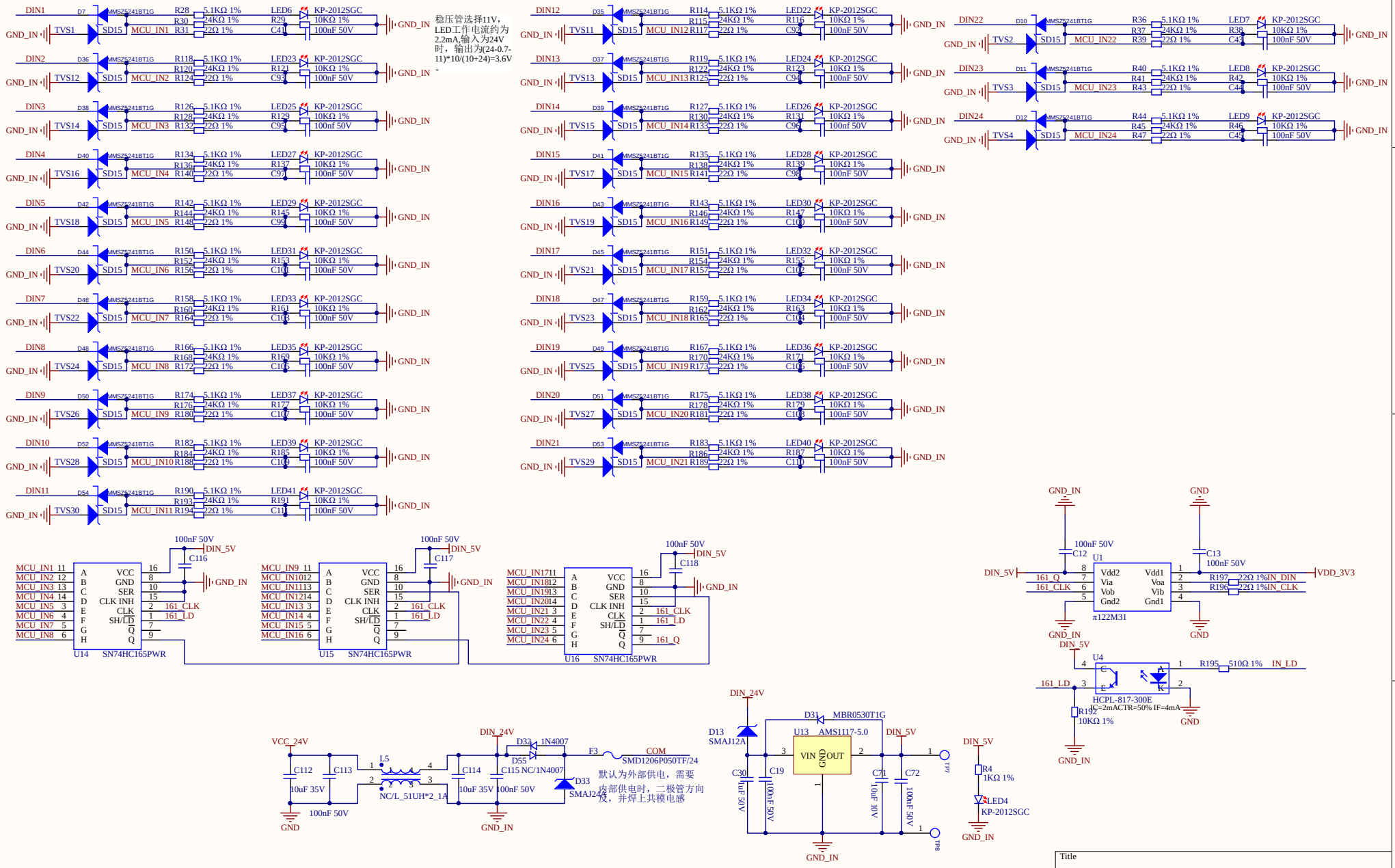


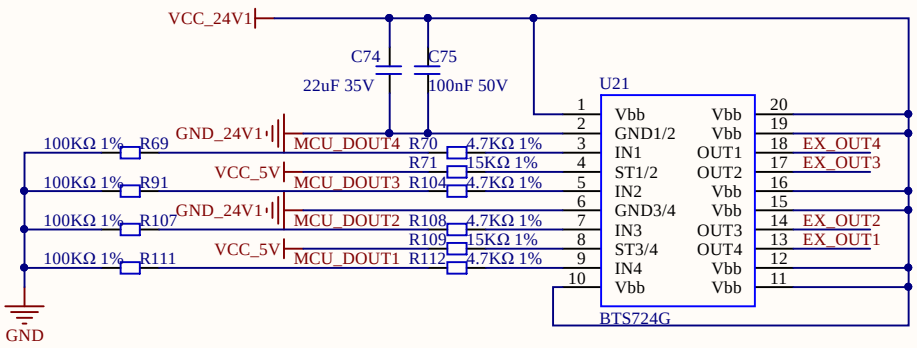



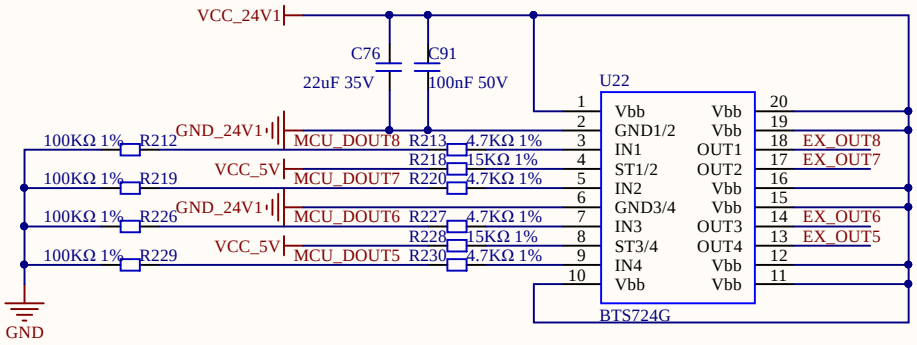

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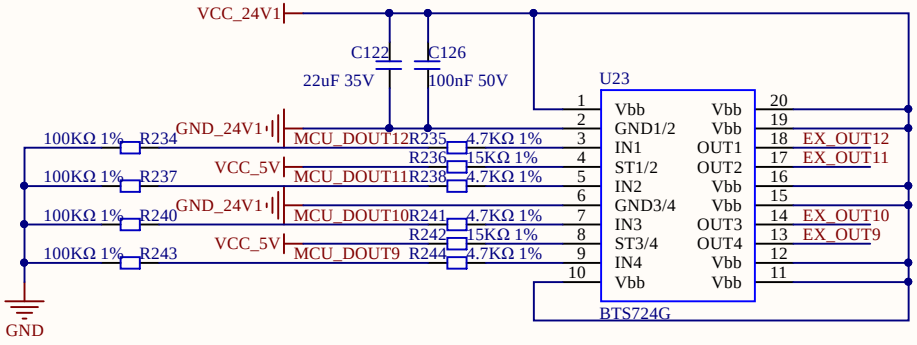


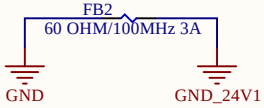

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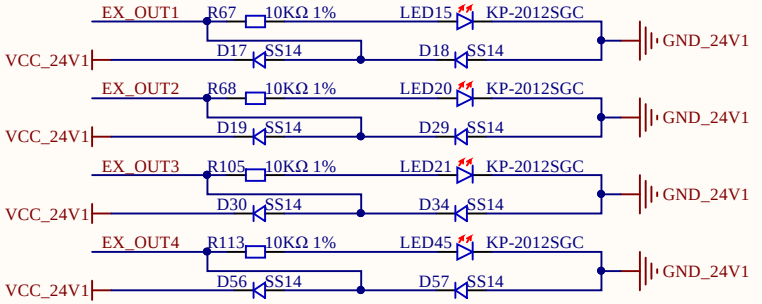

稳压管选择11V，LED工作电流约为2.2mA,输入为24V时，输出为(24-0.7-11)*10/(10+24)=3.6V 。
DIN1 D7 SMMSZ5241BT1G R28 5.1KΩ 1% LED6 KP-2012SGC
GND_IN TVS1 SD15 MCU_IN1 R30 24KΩ 1% R29 10KΩ 1% GND_IN
R31 22Ω 1% C41 100nF 50V
DIN2 D36 R118 5.1KΩ 1% LED23 KP-2012SGC
GND_IN TVS12 SD15 MCU_IN2 R120 24KΩ 1% R121 10KΩ 1% GND_IN
R124 22Ω 1% C93 100nF 50V
DIN3 D38 R126 5.1KΩ 1% LED25 KP-2012SGC
GND_IN TVS14 SD15 MCU_IN3 R128 24KΩ 1% R129 10KΩ 1% GND_IN
R132 22Ω 1% C95 100nf 50V
DIN4 D40 R134 5.1KΩ 1% LED27 KP-2012SGC
GND_IN TVS16 SD15 MCU_IN4 R136 24KΩ 1% R137 10KΩ 1% GND_IN
R140 22Ω 1% C97 100nF 50V
DIN5 D42 R142 5.1KΩ 1% LED29 KP-2012SGC
GND_IN TVS18 SD15 MCU_IN5 R144 24KΩ 1% R145 10KΩ 1% GND_IN
R148 22Ω 1% C99 100nF 50V
DIN6 D44 R150 5.1KΩ 1% LED31 KP-2012SGC
GND_IN TVS20 SD15 MCU_IN6 R152 24KΩ 1% R153 10KΩ 1% GND_IN
R156 22Ω 1% C101 100nF 50V
DIN7 D46 R158 5.1KΩ 1% LED33 KP-2012SGC
GND_IN TVS22 SD15 MCU_IN7 R160 24KΩ 1% R161 10KΩ 1% GND_IN
R164 22Ω 1% C103 100nF 50V
DIN8 D48 R166 5.1KΩ 1% LED35 KP-2012SGC
GND_IN TVS24 SD15 MCU_IN8 R168 24KΩ 1% R169 10KΩ 1% GND_IN
R172 22Ω 1% C105 100nF 50V
DIN9 D50 R174 5.1KΩ 1% LED37 KP-2012SGC
GND_IN TVS26 SD15 MCU_IN9 R176 24KΩ 1% R177 10KΩ 1% GND_IN
R180 22Ω 1% C107 100nF 50V
DIN10 D52 R182 5.1KΩ 1% LED39 KP-2012SGC
GND_IN TVS28 SD15 MCU_IN10 R184 24KΩ 1% R185 10KΩ 1% GND_IN
R188 22Ω 1% C109 100nF 50V
DIN11 D54 R190 5.1KΩ 1% LED41 KP-2012SGC
GND_IN TVS30 SD15 MCU_IN11 R193 24KΩ 1% R191 10KΩ 1% GND_IN
R194 22Ω 1% C111 100nF 50V
DIN12 D35 R114 5.1KΩ 1% LED22 KP-2012SGC
GND_IN TVS11 SD15 MCU_IN12 R115 24KΩ 1% R116 10KΩ 1% GND_IN
R117 22Ω 1% C92 100nF 50V
DIN13 D37 R119 5.1KΩ 1% LED24 KP-2012SGC
GND_IN TVS13 SD15 MCU_IN13 R122 24KΩ 1% R123 10KΩ 1% GND_IN
R125 22Ω 1% C94 100nF 50V
DIN14 D39 R127 5.1KΩ 1% LED26 KP-2012SGC
GND_IN TVS15 SD15 MCU_IN14 R130 24KΩ 1% R131 10KΩ 1% GND_IN
R133 22Ω 1% C96 100nf 50V
DIN15 D41 R135 5.1KΩ 1% LED28 KP-2012SGC
GND_IN TVS17 SD15 MCU_IN15 R138 24KΩ 1% R139 10KΩ 1% GND_IN
R141 22Ω 1% C98 100nF 50V
DIN16 D43 R143 5.1KΩ 1% LED30 KP-2012SGC
GND_IN TVS19 SD15 MCU_IN16 R146 24KΩ 1% R147 10KΩ 1% GND_IN
R149 22Ω 1% C100 100nF 50V
DIN17 D45 R151 5.1KΩ 1% LED32 KP-2012SGC
GND_IN TVS21 SD15 MCU_IN17 R154 24KΩ 1% R155 10KΩ 1% GND_IN
R157 22Ω 1% C102 100nF 50V
DIN18 D47 R159 5.1KΩ 1% LED34 KP-2012SGC
GND_IN TVS23 SD15 MCU_IN18 R162 24KΩ 1% R163 10KΩ 1% GND_IN
R165 22Ω 1% C104 100nF 50V
DIN19 D49 R167 5.1KΩ 1% LED36 KP-2012SGC
GND_IN TVS25 SD15 MCU_IN19 R170 24KΩ 1% R171 10KΩ 1% GND_IN
R173 22Ω 1% C106 100nF 50V
DIN20 D51 R175 5.1KΩ 1% LED38 KP-2012SGC
GND_IN TVS27 SD15 MCU_IN20 R178 24KΩ 1% R179 10KΩ 1% GND_IN
R181 22Ω 1% C108 100nF 50V
DIN21 D53 R183 5.1KΩ 1% LED40 KP-2012SGC
GND_IN TVS29 SD15 MCU_IN21 R186 24KΩ 1% R187 10KΩ 1% GND_IN
R189 22Ω 1% C110 100nF 50V
DIN22 D10 R36 5.1KΩ 1% LED7 KP-2012SGC
GND_IN TVS2 SD15 MCU_IN22 R37 24KΩ 1% R38 10KΩ 1% GND_IN
R39 22Ω 1% C43 100nF 50V
DIN23 D11 R40 5.1KΩ 1% LED8 KP-2012SGC
GND_IN TVS3 SD15 MCU_IN23 R41 24KΩ 1% R42 10KΩ 1% GND_IN
R43 22Ω 1% C44 100nF 50V
DIN24 D12 R44 5.1KΩ 1% LED9 KP-2012SGC
GND_IN TVS4 SD15 MCU_IN24 R45 24KΩ 1% R46 10KΩ 1% GND_IN
R47 22Ω 1% C45 100nF 50V
100nF 50V C116 DIN_5V GND_IN
MCU_IN1 MCU_IN2 MCU_IN3 MCU_IN4 MCU_IN5 MCU_IN6 MCU_IN7 MCU_IN8
A B C D E F G H VCC GND SER CLK INH CLK SH/LD Q Q
161_CLK 161_LD
U14 SN74HC165PWR
100nF 50V C117 DIN_5V GND_IN
MCU_IN9 MCU_IN10 MCU_IN11 MCU_IN12 MCU_IN13 MCU_IN14 MCU_IN15 MCU_IN16
161_CLK 161_LD
U15 SN74HC165PWR
100nF 50V C118 DIN_5V GND_IN
MCU_IN17 MCU_IN18 MCU_IN19 MCU_IN20 MCU_IN21 MCU_IN22 MCU_IN23 MCU_IN24
161_CLK 161_LD 161_Q
U16 SN74HC165PWR
GND_IN GND
100nF 50V C12 C13 100nF 50V
U1 Vdd2 Via Vob Gnd2 Vdd1 Voa Vib Gnd1
DIN_5V 161_Q 161_CLK
R197 22Ω 1% IN_DIN
R196 22Ω 1% IN_CLK
VDD_3V3
π122M31
U4 HCPL-817-300E
CTR=2mACTR=50% IF=4mA
R195 510Ω 1% IN_LD
161_LD
R194 10KΩ 1%
VCC_24V C112 10uF 35V C113 100nF 50V GND
L5 NC/L_51UH*2_1A
DIN_24V D32 1N4007 D55 C114 10uF 35V C115 NC/1N4007 100nF 50V GND_IN
D33 SMAJ24A
F3 SMD1206P050TF/24
默认为外部供电，需要内部供电时，二极管方向反，并焊上共模电感
DIN_24V D13 SMAJ12A D31 MBR0530T1G
U13 AMS1117-5.0 VIN GND OUT
C30 10uF 50V C19 100nF 50V C71 10uF 10V C72 100nF 50V
DIN_5V TP7 TP8 GND_IN
DIN_5V R4 1KΩ 1% LED4 KP-2012SGC GND_IN
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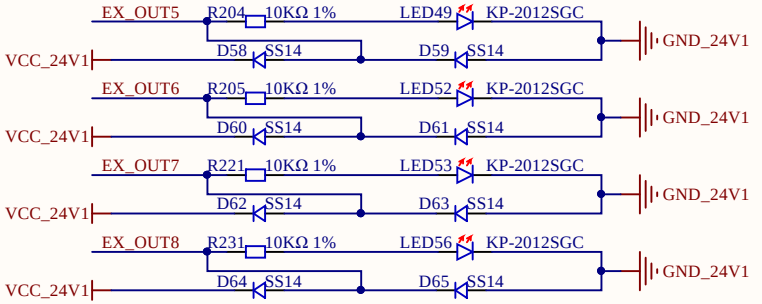


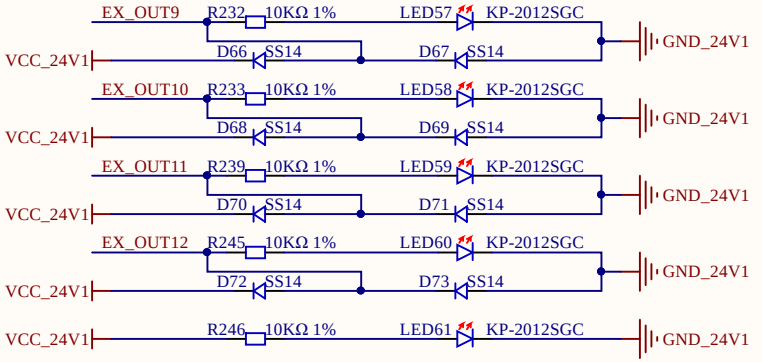



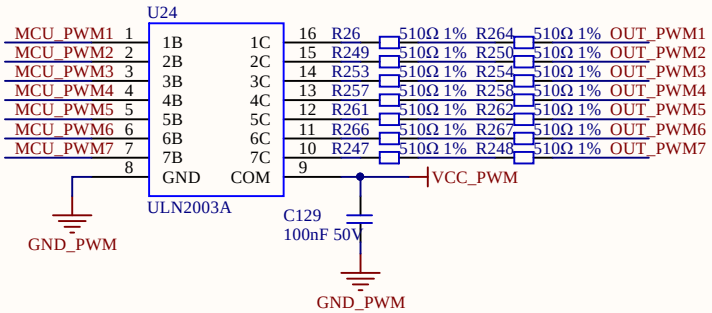



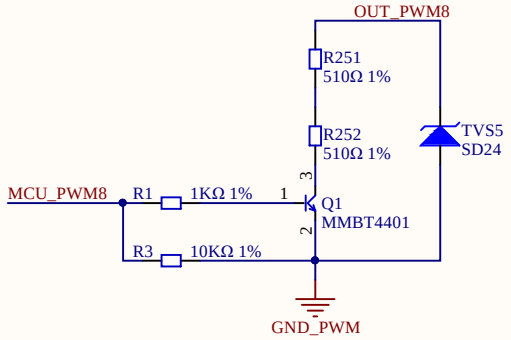



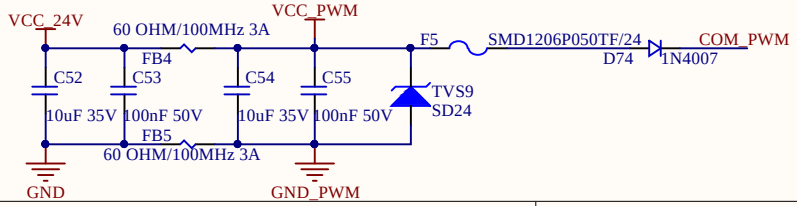



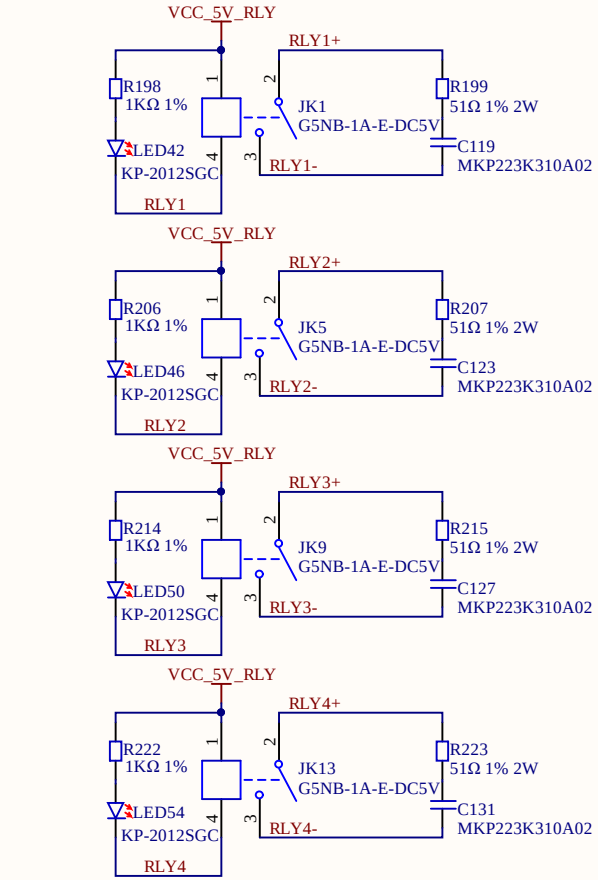

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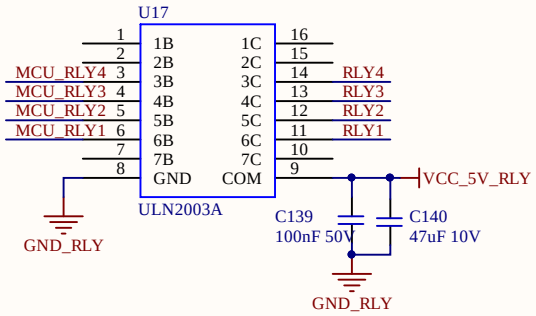


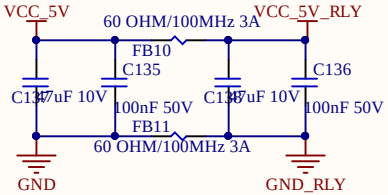



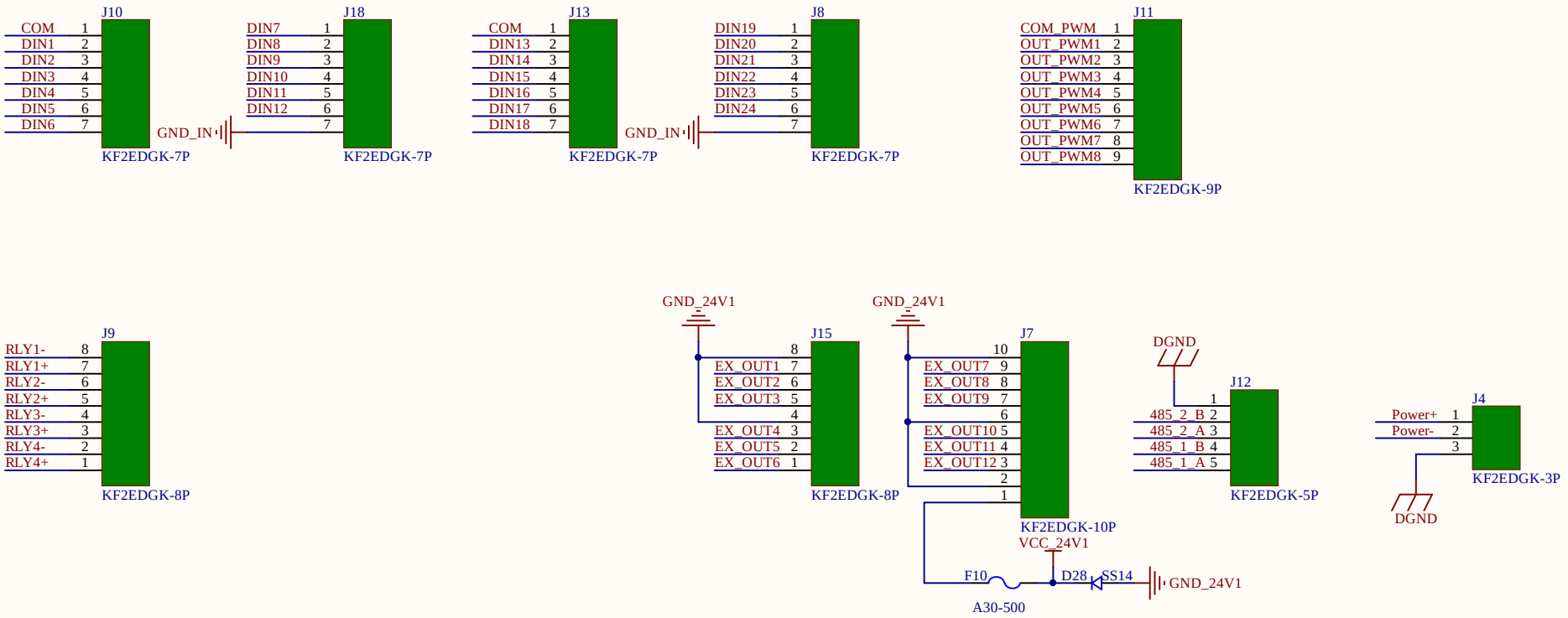

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DIN7 DIN8 DIN9 DIN10 DIN11 DIN12 GND
COM DIN13 DIN14 DIN15 DIN16 DIN17 DIN18
DIN19 DIN20 DIN21 DIN22 DIN23 DIN24 COM
PWM1 PWM2 PWM3 PWM4 PWM5 PWM6 PWM7 PWM8
3.3V
SWD
GND
CLK
RES
ZNMC_612_BZTDKZ1.1..
2023H
GND
TX
RX
RLY1 RLY2 RLY3 RLY4
PMOS-OUT
GND 1 2 3 GND 4 5 6
PMOS-OUT
GND 7 8 9 GND 10 11 12
Power
GND 24V
RS485-1
A B
RS485-2
A B
PE
PE
Power
GND 24V